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Sofia de Oliveira, Azucena López-Muñoz, Sergio Candel, Pablo Pelegrín, Ángel Calado and Victoriano Mulero

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ATP Modulates Acute Inflammation In Vivo through Dual Oxidase 1–Derived H₂O₂ Production and NF-κB Activation

Sofia de Oliveira,^{*,†,‡} Azucena López-Muñoz,^{‡,§} Sergio Candel,^{‡,§} Pablo Pelegrín,^{§,¶} Ângelo Calado,^{*,†} and Victoriano Mulero^{‡,§}

Dual oxidase 1 (Duox1) is the NADPH oxidase responsible for the H₂O₂ gradient formed in tissues after injury to trigger the early recruitment of leukocytes. Little is known about the signals that modulate H₂O₂ release from DUOX1 and whether the H₂O₂ gradient can orchestrate the inflammatory response in vivo. In this study, we report on a dominant-negative form of zebrafish Duox1 that is able to inhibit endogenous Duox1 activity, H₂O₂ release and leukocyte recruitment after tissue injury, with none of the side effects associated with morpholino-mediated Duox1 knockdown. Using this specific tool, we found that ATP release following tissue injury activates purinergic P2Y receptors, and modulates Duox1 activity through phospholipase C (PLC) and intracellular calcium signaling in vivo. Furthermore, Duox1-derived H₂O₂ is able to trigger the NF-κB inflammatory signaling pathway. These data reveal that extracellular ATP acting as an early danger signal is responsible for the activation of Duox1 via a P2YR/PLC/Ca²⁺ signaling pathway and the production of H₂O₂, which, in turn, is able to modulate in vivo not only the early recruitment of leukocytes to the wound but also the inflammatory response through activation of the NF-κB signaling pathway. *The Journal of Immunology*, 2014, 192: 5710–5719.

Dual oxidase 1 (DUOX1), an NADPH oxidase (NOX) family member (1–4), is responsible for the formation of hydrogen peroxide (H₂O₂) tissue gradient formed after wounding (5–7). This gradient has been shown to be the main trigger of early leukocyte recruitment and is later required for tissue regeneration via oxidation of the Src family kinases (SFKs), Lyn and Fyn b, respectively (6–8). Importantly, neutrophil-delivered myeloperoxidase is responsible for the rapid clearance of H₂O₂ after injury (9), preventing excessive tissue damage. In addition, DUOX1-derived H₂O₂ is essential in host defense by supporting lactoperoxidase-

mediated antimicrobial defense mechanisms on mucosal surfaces (10). Despite our increased knowledge about H₂O₂ signaling, the mechanisms that regulate in vivo DUOX1 activation are poorly understood. In the injured epidermis of *Drosophila* embryos, it has recently been reported that calcium flashes orchestrate the wound inflammatory response through DUOX1 activation, via its EF hand calcium-binding motif (7), whereas in zebrafish this does not seem to be the case (8). As in *Drosophila* (7), the zebrafish Duox1 (5, 7) also has two canonical EF hands in an intracellular loop, which suggests that cytosolic calcium might regulate H₂O₂ production. However, to date, little is known about the role of cytosolic calcium signaling in the modulation of DUOX1 activation and H₂O₂ production in vivo. Besides the generation of calcium waves (11–13), the release of ATP is an important early danger signal after tissue damage (14–16). In addition, the activation of purinergic receptors by extracellular ATP is one of the main biological mechanisms responsible for epithelial intracellular calcium (iCa²⁺) mobilization (16–19) and several in vitro studies have correlated this signal with DUOX1 activation (20–24).

Redox processes regulate different oxygen-sensitive transcription factors through the direct modification of proteins (25). Altering gene expression is a fundamental mechanism for cells to respond to changes in the extracellular environment. NF-κB is an important redox-sensitive transcription factor (25, 26), which regulates the expression of the genes involved in multiple biological functions, including inflammation, apoptosis and proliferation (27). However, there is controversy concerning whether H₂O₂ activates or inhibits NF-κB (28, 29). Furthermore, there is increasing evidence that H₂O₂ is not just a generic modulator of transcription factors and signaling molecules but also may act as a specific regulator of individual genes (29–33).

In this study, we investigate whether ATP release acts as a danger signal to mobilize intracellular calcium, acting upstream of DUOX1 activation in vivo and further modulating NF-κB activation in acute inflammatory conditions. We show that zebrafish Duox1 activation and subsequent neutrophil recruitment are mediated by ATP acting through a P2Y receptor (P2YR)/phospholipase

*Laboratório de Carlota Saldanha, Instituto de Medicina Molecular, 1649-028 Lisbon, Portugal; †Instituto de Bioquímica, Faculdade de Medicina, Universidade de Lisboa, 1649-028 Lisbon, Portugal; ‡Departamento de Biología Celular e Histología, Facultad de Biología, Universidad de Murcia, 30100 Murcia, Spain; §Instituto Murciano de Investigación Biosanitaria, 30120 Murcia, Spain; and ¶Unidad de Inflamación y Cirugía Experimental, Centro de Investigación Biomédica en Red en el Área Temática de Enfermedades Hepáticas y Digestivas, Hospital Universitario Virgen de la Arrixaca, 30120 Murcia, Spain

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S.d.O. and V.M. conceived the study; S.d.O., P.P., A.C., and V.M. designed research; S.d.O. and A.L.-M. performed research; S.d.O., A.L.-M., S.C., P.P., A.C., and V.M. analyzed data; and S.d.O. wrote the manuscript with contributions from other authors.

Address correspondence and reprint requests to Prof. Victoriano Mulero, Departamento de Biología Celular e Histología, Facultad de Biología, Universidad de Murcia, 30100 Murcia, Spain. E-mail address: vmulero@um.es

The online version of this article contains supplemental material.

Abbreviations used in this article: CHT, caudal haematopoietic tissue; DN, dominant-negative; dpf, d postinfection; DPI, dibenzodioxolium chloride; Duox, dual oxidase; FAD, flavin adenine dinucleotide; iCa²⁺, intracellular calcium; LTB₄, leukotriene B₄; minpi, min postinjection; minpw, minutes postwounding; MO, morpholino; NOX, NADPH oxidase; PLC, phospholipase C; PPADS, pyridoxal phosphate-6-azo(benzene-2,4-disulfonic acid) tetrasodium salt hydrate; P2YR, purinergic 2Y receptor; SERCA, sarco/endoplasmic reticulum calcium ATPase; TG, thapsigargin.

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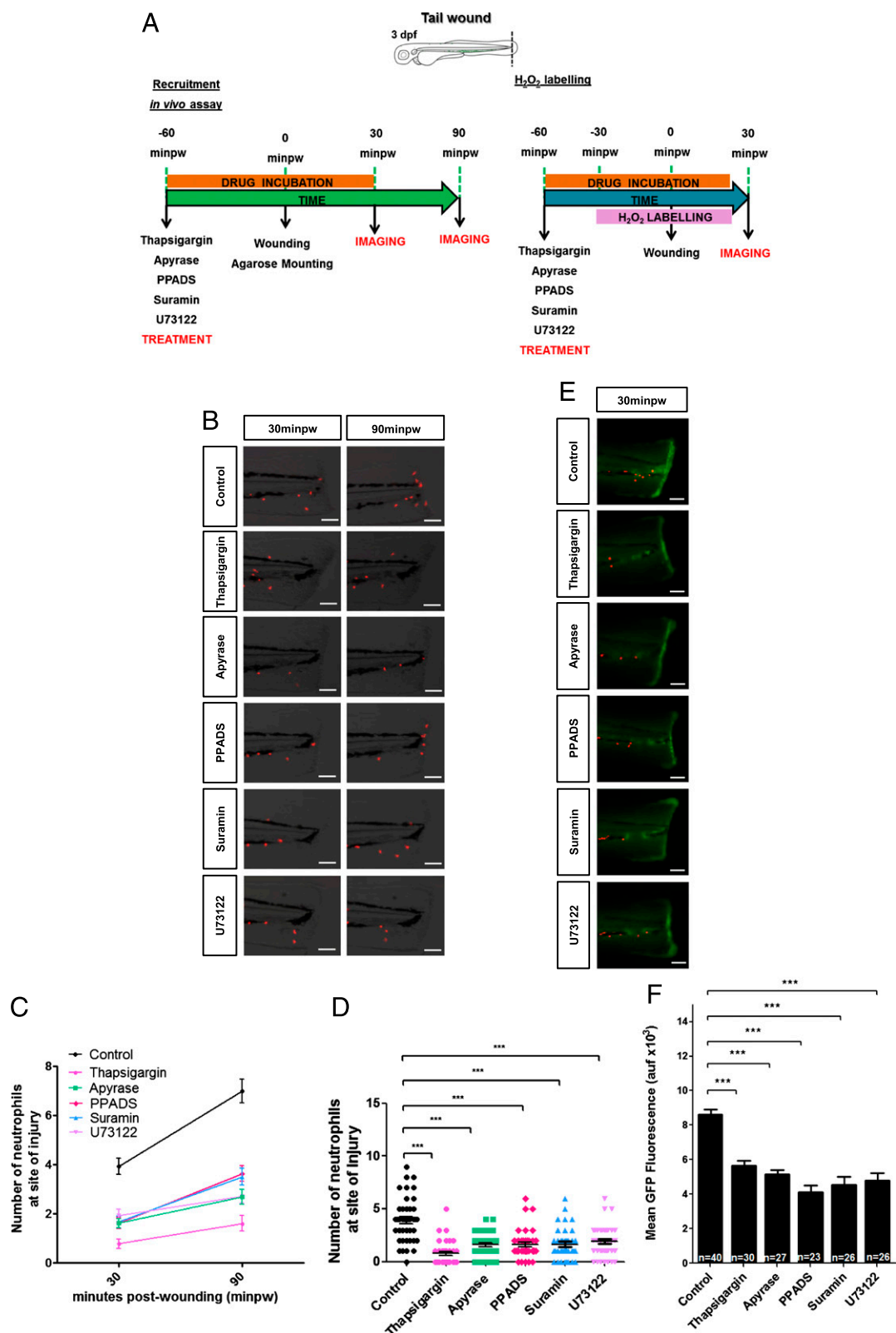


FIGURE 1. ATP and calcium signaling modulate Duox1-derived H₂O₂ release and neutrophil recruitment in wounding. **(A)** Schematic representation of the neutrophil recruitment and H₂O₂ visualization assays. **(B–F)** Uninjected 3 dpf Tg(lyz:DsRED2)^{nz50} larvae were treated with 10 U/ml apyrase, 1 μM thapsigargin (TG), 100 μM PPADS, 100 μM suramin, or 1 μM U73122, and further tail fin larvae amputation was performed. For H₂O₂ labeling, larvae were incubated for 30 min with 50 mM acetyl-pentafluorobenzene sulphonyl fluorescein in 1% DMSO in embryo medium. **(B)** Representative overlay images of bright-field and green channels of wounded larvae tail fins at 30 and 90 minpw. **(C and D)** Counts of fluorescent neutrophils at the site of injury were made 30 minpw (**C** and **D**) and 90 minpw (**C**). (Control: *n* = 33, apyrase: *n* = 36, TG: *n* = 35, PPADS: *n* = 37, suramin: *n* = 38, U73122: *n* = 36). All treatments were able to significantly decrease neutrophil recruitment to wounds. Representative maximum intensity projections of H₂O₂ production of red (neutrophils) and green (H₂O₂) channels (**E**) and wound fluorescence intensity quantification in wounded larvae tail fins at 30minpw labeled with acetyl-pentafluorobenzene sulphonyl fluorescein (**F**). Control: *n* = 27, apyrase: *n* = 25, TG: *n* = 28, PPADS: *n* = 22, suramin: *n* = 22, (Figure legend continues)

C (PLC)/calcium signaling pathway and that Duox1-derived H_2O_2 activates in vivo the NF- κ B inflammatory signaling pathway.

Materials and Methods

Zebrafish husbandry

All experiments with live animals were performed using protocols approved by the European Union Council Guidelines (86/609/EU) and the Bioethical Committee of the University of Murcia (approval number 537/2011). Fertilized zebrafish eggs were obtained from natural spawning of wild-type (obtained from the Zebrafish International Resource Center), the Tg(mpx:gfp)ⁱ¹¹⁴ (34) and the Tg(lyz:DsRED2)^{nz50} (35) lines held at our facilities following standard husbandry practices. Tg(NF- κ B-RE:eGFP) (NF- κ B:eGFP for even greater simplicity) line was generated with the method and constructs as described previously (36). Animals were maintained in a 12-h light/dark cycle at 28.5°C.

DNA constructs

Zebrafish Duox1 amino acid sequence (XP_001919394.3) was submitted to domain analysis using the Pfam database (<http://pfam.sanger.ac.uk/>) (37). A truncated form of the wild-type Duox1, which lacks the entire flavin domain (residues 1–1232, dominant-negative [DN] Duox1; Supplemental Fig. 1A), was chemically synthesized and cloned in pBluescript II KS+ (GenScript). CMV/T7:DN-Duox1:GFP was generated by MultiSite Gateway assemblies using LR Clonase II Plus (Life Technologies), according to standard protocols and using the Tol2kit vectors as described previously (38).

Morpholino and mRNA injections

Specific morpholinos (MOs; Gene Tools) were resuspended in nuclease-free water. In vitro-transcribed DN-Duox1 (sense and antisense), DN-Duox1:GFP, and GFP RNA were obtained following the manufacturer's instructions (mMESSAGE mMACHINE kit; Ambion). RNAs (300 pg/egg) and duox1 splice MO (5'-AGTGAATTAGAGAAATGCACCTTTT-3') (125 μ M) with p53 MO (5'-GCGCCATTGCTTTGCAAGAATTG-3') (100 μ M) (5, 6) were mixed in microinjection buffer (0.5 $\times$ Tango buffer and 0.05% phenol red solution) and microinjected into one-cell stage embryos using a microinjector (Narishige) (0.5–1 nl/embryo). The same amounts of RNA and MOs were used in all experimental groups.

HEK293 cell transfection and Western blot

Plasmid DNAs were prepared using the Midi-Prep procedure (Qiagen) and transfected into HEK293 cells with LyoVec transfection reagent (InvivoGen), according to the manufacturer's instructions. At 48 h after transfection, the cells were washed twice with PBS and lysed in 200 μ l lysis buffer (10 mM Tris-HCl [pH 7.4] and 1% SDS). The protein concentrations of cell lysates were estimated by the bicinchoninic acid protein assay reagent (Pierce) using BSA as a standard. Cell extracts (50 μ g protein) were analyzed on 8% (DN-Duox1:GFP) and 15% (GFP) SDS-PAGE and transferred for 50 min at 200 mA to nitrocellulose membranes (Bio-Rad). The blots were developed with mouse anti-GFP Ab (1/1000) (BD Clontech) and ECL reagents (GE Healthcare), according to the manufacturer's protocol.

Tail fin wounding

Essentially, tail fin amputation was performed as described previously (39). Briefly, Tg(lyz:DsRED2)^{nz50}, Tg(mpx:gfp)ⁱ¹¹⁴, or Tg(NF- κ B:EGFP) larvae were anesthetized at 3 d postinfection (dpf) in embryo medium with 0.16 mg/ml tricaine. Then, complete transection of the tailfin tip was performed with a disposable sterile scalpel, and fish were mounted in 1% (w/v) low-melting point agarose (Sigma-Aldrich) dissolved in embryo medium supplemented with 0.16 mg/ml tricaine. The success of transection was immediately confirmed by in a fluorescence stereo microscope MZ16FA (Leica) equipped with green and red fluorescent filters. After solidification, embryo medium with 0.16 mg/ml tricaine solution, pre-treated or not with pharmacological inhibitors (see below), was added to keep the embryos hydrated during experiments. Then, images were captured at the selected times while animals were kept in their agar matrixes at 28.5°C.

Pharmacological treatment

All drug treatments were made using the bath immersion method. Briefly, 3 dpf Tg(lyz:DsRED2)^{nz50}, Tg(mpx:gfp)ⁱ¹¹⁴, or Tg(NF- κ B:EGFP) larvae were incubated for 1 h at 28°C in the presence or absence of each of the following drugs: 100 μ M dibenzodiolium chloride (DPI), 1 μ M thapsigargin (TG), 1 μ M U73122, 100 μ M pyridoxal phosphate-6-azo(benzene-2,4-disulfonic acid) tetrasodium salt hydrate (PPADS), 100 μ M suramin, and 10 U/ml apyrase (all from Sigma-Aldrich) diluted in embryo medium supplemented (or not) with 1% DMSO. After tail fin amputation or otic injection, the larvae were kept in the corresponding treatments until imaging.

H₂O₂ imaging

H₂O₂ imaging using a live cell fluorescein dye was performed as described previously (40). Briefly, 3 dpf Tg(lyz:DsRED2)^{nz50} larvae pretreated with RNA, MOs or drugs and corresponding control siblings, were loaded for 30 min with 50 mM acetyl-pentafluorobenzene sulphonyl fluorescein (Cayman Chemical) in 1% DMSO in embryo medium, prior to tail fin amputation, carried out as described above. Larvae were left to recover in probe solution and imaging was made 30 min postwounding (minpw). The same number of larvae were left without wounding in each group/experiment and used as fluorescence background controls for mean fluorescence intensity quantification.

Otic injection

Otic vesicle injections of 3 dpf Tg(lyz:DsRED2)^{nz50} larvae for different treatments were performed as previously described (39), 1 nl 1 $\times$ PBS, 10 μ M H₂O₂, 100 μ M ATP, 100 μ M ADP, 1 μ M A23187 (Sigma-Aldrich), or 30 nM leukotriene B₄ (LTB₄) (Cayman Chemical). Images were taken 30 min postinjection (minpi).

Image acquisition and processing

For each experiment, 3 dpf morphant and control larvae were imaged in three independent experiments. Images were taken from wounded or control larvae mounted as described above. Briefly, for total neutrophil and neutrophils at the site of injury/injection counts, images were taken using a Leica MZ16F fluorescence stereo microscope equipped with green and red fluorescent filters, while the animals were maintained in their agar matrixes at 28.5°C. Stacked images were captured using 20 μ m (neutrophil distribution, NF- κ B activation, and H₂O₂ formation) increments and deconvolved using Huygens Essential Confocal software (version 4.1.0p6b) by Scientific Volume Imaging. Stacks were then processed using the free source software ImageJ (<http://rsbweb.nih.gov/ij/>) to obtain a maximum intensity projection of the xy axis, and mean fluorescence at the wound site was again quantified for each experiment. Mean fluorescence was measured in wounded epithelial cells and the corresponding background values of unwounded larvae tail fins were subtracted for experiments of NF- κ B activation (same larvae were used at 0 and 30 minpw) and H₂O₂ formation (same number of larvae wounded and unwounded were imaged 30 minpw with the same time of probe incubation).

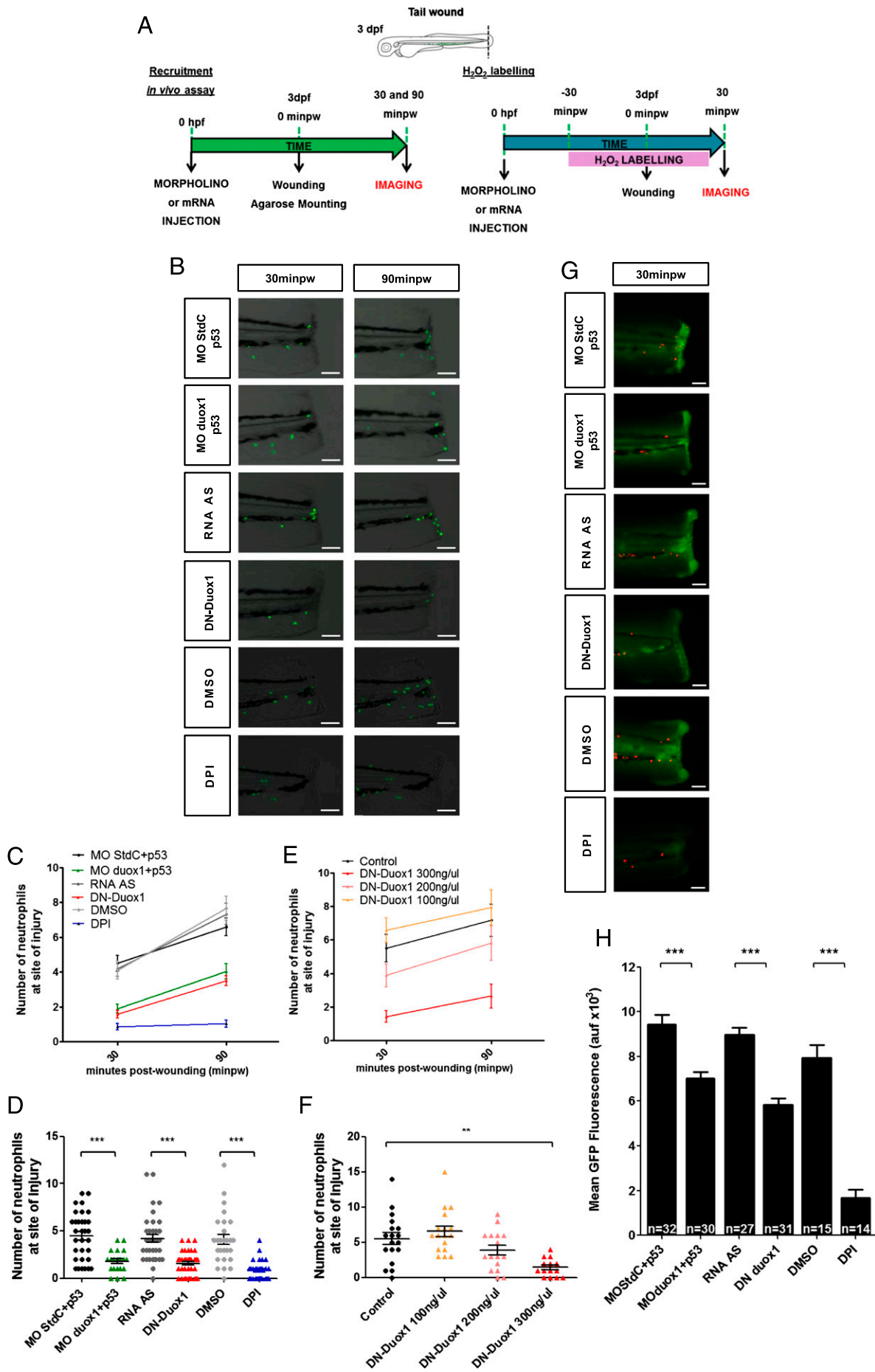
Statistical analysis

All error bars indicates SEM. A one-way ANOVA with Bonferroni posttest was used.

Results

ATP and calcium signaling modulate Duox1-derived H₂O₂ production and neutrophil recruitment in wounding

We first focused on the signals that might be responsible for the activation/modulation of Duox1 in vivo. Extracellular ATP and Ca²⁺ appeared to be the strongest candidates to exert this function (11–14). Therefore, apyrase was used to degrade extracellular ATP (20) and TG, a sarco/endoplasmic reticulum calcium ATPase inhibitor (8), to block Ca²⁺ signaling (Fig. 1A). It was seen that the inhibition of ATP or Ca²⁺ signaling significantly reduced early neutrophil recruitment to the wound 30 and 90 min postwounding



(minpw) (Fig. 1B–D) and also H_2O_2 production, assayed by the oxidation of the highly H_2O_2 -specific probe acetyl-pentafluorobenzene sulphonyl fluorescein (5, 40), in tail fin wounded tissues 30 minpw (Fig. 1E, 1F).

To further investigate the signaling pathway involved in the activation of Duox1 by ATP and Ca^{2+} , two general pharmacological inhibitors of purinergic receptors, suramin and PPADS (17, 20), and the PLC inhibitor U73122 (8), were used (Fig. 1A). The resulting data show that the inhibition of purinergic signaling and PLC significantly reduced early neutrophil recruitment to the wound 30 and 90 minpw (Fig. 1B–D) and H_2O_2 production in tail fin wounded tissues 30 minpw (Fig. 1E, 1F).

A Duox1 mutant lacking the flavin domain acts as a DN in vivo

Duox1 splice blocking MOs have been widely used to inhibit Duox1 in zebrafish, but they need to be tightly controlled and coinjected with a MO against p53 because of widespread cell death and developmental delay (5, 6, 41). For this reason, we developed a dominant negative form of Duox1 (DN-Duox1) lacking the carboxyl-terminal flavin domain, which contains the binding sites for the cofactors NADPH and flavin adenine dinucleotide (FAD) (Supplemental Fig. 1A), as has been reported for NOX4 (42). The DN-Duox1 mRNA or the Duox1 MO was microinjected into one-cell stage embryos and the total number of neutrophils and their development were assessed 3 d postfertilization (Supplemental Fig. 1B, 1C). As reported before (5, 6, 43), the Duox1 morphant larvae were smaller and showed delayed development compared with larvae in control conditions or larvae microinjected with DN-Duox1 mRNA (Supplemental Fig. 1B). However, neither the Duox1 MO nor the DN affected total neutrophil numbers (Supplemental Fig. 1C). We noticed that Duox1 morphants presented an abnormal neutrophil distribution, so we quantified the number of neutrophils present above the notochord and in the caudal hematopoietic tissue (CHT), where most neutrophils are located at this developmental stage (44, 45), and calculated the percentage of neutrophils above the notochord. It was found that Duox1 MO significantly increased the percentage of neutrophils above notochord; that is, outside of the CHT, whereas the injection of DN-Duox1 mRNA did not (Supplemental Fig. 1D).

Next, it was checked whether the DN-Duox1 was able to reduce neutrophil recruitment toward a wound and H_2O_2 production and release. For this, we performed in vivo neutrophil recruitment assays in 3 dpf larvae (Fig. 2A), using tail fin amputation as an acute inflammatory stimulus and DPI, a NADPH oxidase general inhibitor (5, 46), as a control. As expected, larvae microinjected with DN-Duox1 mRNA had a significantly lower number of neutrophils at the site of injury 30 and 90 minpw (Fig. 2B–D) and

showed the same level of recruitment as the morphants. In addition, the effect of the DN seemed to be specific because it was dose dependent (Fig. 2E, 2F). Also, we observed that the DN-Duox1 RNA and the MO were both able to significantly reduce the amount of H_2O_2 produced by wounded tissue 30 minpw (Fig. 2G, 2H). Moreover, the effect of DN-Duox1 on H_2O_2 -mediated neutrophil recruitment was specific, because it did not affect neutrophil recruitment to other stimuli, such as LTB₄ (Supplemental Fig. 2A).

To verify the expression of DN-Duox1, a GFP-tagged DN-Duox1 form was generated. The mRNA encoding DN-Duox1:GFP was microinjected into one cell stage embryos and neutrophil numbers at the injury site after tail fin amputation were quantified at 3 dpf (Supplemental Fig. 2B, 2C). Although these larvae presented a significant decrease in neutrophil recruitment and similar to untagged DN-Duox1 (Supplemental Fig. 2B–D), we were unable to detect a DN-Duox1:GFP signal by fluorescence microscopy (data not shown). However, we were able to detect GFP signal in HEK293 cells transfected with the same DN-Duox1:GFP construct driven by the CMV promoter (data not shown). Western blot analysis of HEK293 transfected with the DN-Duox1:GFP construct confirmed the expression of the chimeric protein with the expected m.w. (Supplemental Fig. 2E).

ATP modulates Duox1 activity through calcium signaling in vivo

At this point, we have shown that purinergic ATP signaling, probably through intracellular calcium rise, is able to modulate H_2O_2 production and neutrophil recruitment. However, whether these signals act in parallel or whether one is upstream of the other is still unknown. In an attempt to resolve this issue, we microinjected the DN-Duox1 mRNA into one-cell stage embryos and pretreated them 3 dpf with apyrase, TG and U73122 for 60 min, before being microinjected into the otic vesicle with different stimuli, including H_2O_2 , ATP, ADP, and the Ca^{2+} ionophore A23187 (Fig. 3A). In control conditions, all stimuli significantly increased neutrophil recruitment to the ear cavity of larvae, but, in the presence of the DN-Duox1 form, only H_2O_2 was able to significantly increase neutrophil recruitment (Fig. 3B, 3C), suggesting that purinergic and Ca^{2+} signaling were both upstream of Duox1 activation and H_2O_2 release. In addition, in the presence of apyrase, TG, or U73122 (used to inhibit ATP, Ca^{2+} , and PLC signaling, respectively) only H_2O_2 and A23187 were able to significantly increase neutrophil recruitment to the ear cavity (Fig. 3B, 3C), indicating that purinergic signaling was upstream of Ca^{2+} , and strongly suggesting the involvement of a P2Y metabotropic receptor that would lead to increased intracellular Ca^{2+} (iCa^{2+}) because of depletion of the intracellular stores through PLC activation.

FIGURE 2. DN-Duox1 inhibits neutrophil recruitment and H_2O_2 release in wounding. **(A)** Schematic representation of the neutrophil recruitment and H_2O_2 labeling assays. **(B–H)** Zebrafish one-cell Tg(mpx:GFP)¹¹⁴ or Tg(lyz:DsRED2)^{nz50} was microinjected with mRNA antisense (RNA AS) or sense (DN-Duox1), standard control MO (MO StdC), or Duox1 splice blocking MO (MO duox1) both coinjected with p53 ATG MO (MO p53). Uninjected larvae were treated 3 dpf with DPI or DMSO and used as controls. Tail fin larvae were amputated at 3 dpf in all groups. For H_2O_2 labeling, larvae were incubated for 30 min with 50 mM acetyl-pentafluorobenzene sulphonyl fluorescein in 1% DMSO in embryo medium. **(B)** Representative overlay images of brightfield and green channels of wounded larvae tail fins 30 and 90 minpw. **(C and D)** Counts of fluorescent neutrophils at the site of injury were made at 30 minpw (C and D) and 90 minpw (C). (MO StdC+p53: $n = 32$, MO duox1+p53: $n = 31$, RNA AS: $n = 34$, DN-Duox1: $n = 37$, DPI: $n = 27$, DMSO: $n = 29$). DPI, DN-Duox1 and Duox1 MO significantly decreased neutrophil recruitment to wounds. **(E and F)** Counts of fluorescent neutrophils at the site of injury were made 30 minpw (E and F) and 90 minpw (E) in larvae microinjected with different doses of mRNA. (RNA AS 300 ng/ μ l: $n = 18$, DN-Duox1 100 ng/ μ l: $n = 17$, DN-Duox1 200 ng/ μ l: $n = 18$, DN-Duox1 300 ng/ μ l: $n = 17$). mRNA of DN-Duox1 significantly decreased neutrophil recruitment to wounds in a dose-dependent manner. Representative maximum intensity projections of H_2O_2 production of red (neutrophils) and green (H_2O_2) channels **(G)** and wound fluorescence intensity quantification of wounded larvae tail fins at 30 minpw labeled with acetyl-pentafluorobenzene sulphonyl fluorescein **(H)**. (MO StdC+p53: $n = 32$, MO duox1+p53: $n = 30$, RNA AS: $n = 27$, DN-Duox1: $n = 31$, DPI: $n = 15$, DMSO: $n = 14$). DPI, DN-Duox1, and Duox1 MO were able to significantly decrease H_2O_2 production in wounded tail fins. All data are represented as means $\pm$ SEM. Scale bars, 100 μ m. The p values were calculated using one-way ANOVA and Bonferroni multiple comparison test (** $p < 0.01$, *** $p < 0.001$). a.u.f., arbitrary unit of fluorescence.

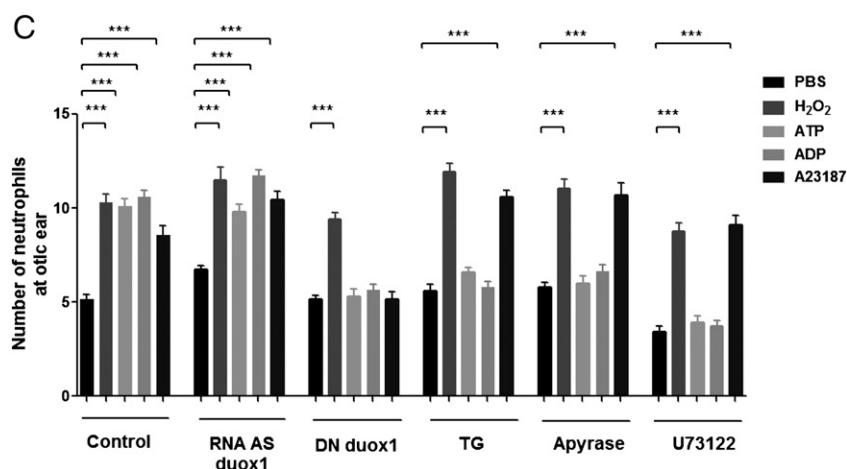
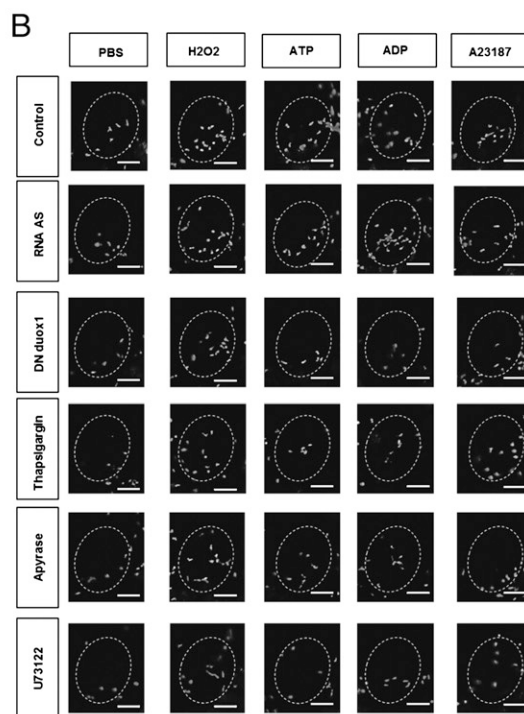
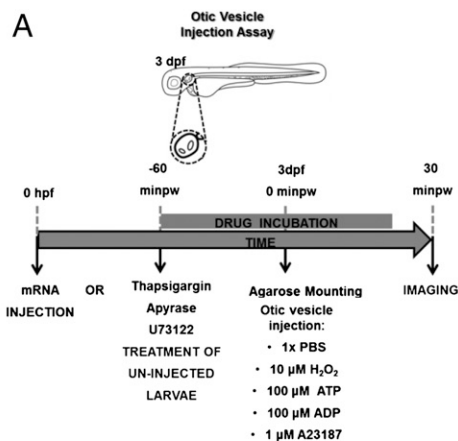


FIGURE 3. ATP modulates Duox1 activity acting through P2Y₂/iCa²⁺ signaling pathway. **(A)** Schematic representation of the otic vesicle injection assay. **(B and C)** Zebrafish one-cell Tg(lyz:DsRED2)^{nz50} were microinjected with anti-sense (RNA AS) or sense (DN-Duox1) DN-Duox1 mRNA. At 3 dpf, uninjected larvae were treated with TG, apyrase, or U73122. Otic vesicle microinjection of 1 nl PBS, 10 μ M H₂O₂, 100 μ M ATP, 100 μ M ADP, and 1 μ M A23187 was performed in each group. **(B)** Representative images of red channel of otic vesicles, 30 minpi. **(C)** Counts of fluorescent neutrophils at the site of injection were made 30 minpi. (Control/PBS: *n* = 90, Control/H₂O₂: *n* = 60, Control/ATP: *n* = 60, Control/ADP: *n* = 72, Control/A23187: *n* = 70, RNA AS/PBS: *n* = 71, RNA AS/H₂O₂: *n* = 38, RNA AS/ATP: *n* = 35, RNA AS/ADP: *n* = 40, RNA AS/A23187: *n* = 40, DN-Duox1/PBS: *n* = 70, DN-Duox1/H₂O₂: *n* = 38, DN-Duox1/ATP: *n* = 40, DN-Duox1/ADP: *n* = 35, DN-Duox1/A23187: *n* = 38, TG/PBS: *n* = 40, TG/H₂O₂: *n* = 40, TG/ATP: *n* = 37, TG/ADP: *n* = 37, TG/A23187: *n* = 40, apyrase/PBS: *n* = 70, apyrase/H₂O₂: *n* = 40, apyrase/ATP: *n* = 37, apyrase/ADP: *n* = 37, apyrase/A23187: *n* = 38, U73122/PBS: *n* = 37, U73122/H₂O₂: *n* = 41, U73122/ATP: *n* = 43, U73122/ADP: *n* = 45, U73122/A23187: *n* = 41). Note that all stimuli significantly increased the number of neutrophils recruited to the ear in control and RNA AS samples, but only H₂O₂ was able to significantly increase neutrophil recruitment to ear in DN-Duox1 larvae. In larvae pretreated with TG, apyrase, or U73122, H₂O₂ and the calcium ionophore A23187 were also the only stimuli able to significantly increase neutrophil recruitment. All data are represented as means $\pm$ SEM. Scale bars, 100 μ m. The *p* values were calculated using one-way ANOVA and Bonferroni multiple comparison test (****p* < 0.001).

H₂O₂ activates the NF-κB inflammatory signaling pathway

NF- κ B is a master regulator of inflammation, playing an essential role in the induction of several proinflammatory genes (47). Thus, we wanted to ascertain whether Duox1-derived H₂O₂ was able to activate this inflammatory signaling pathway *in vivo*. For this,

we used the transgenic zebrafish line that expresses enhanced GFP under transcriptional control of NF- κ B to quantify NF- κ B activation in 3 dpf larvae after wounding in the presence of DN-Duox1, apyrase, or TG. DPI was again used as a positive control for the experiment. As expected, the NF- κ B signaling

pathway was activated in wounded tail fin tissues (Fig. 4A, 4B). In addition, when Duox1 was silenced by the use of the DN form or by inhibiting ATP or Ca^{2+} signaling with drugs, the NF- κ B activation significantly decreased in wounded tissue (Fig. 4A, 4C). These results indicate that the activation of Duox1 by ATP and Ca^{2+} signaling leads to the release of H_2O_2 , which, in turn, promotes the local activation of NF- κ B.

Discussion

During recent years, the H_2O_2 tissue gradient has emerged as a very important process involved in several mechanisms such as leukocyte recruitment (5, 6, 8), tumor progression (48) and neuron regeneration (43). Tissue-scale H_2O_2 gradient and its roles have been studied extensively, although the upstream signals involved in early Duox1 activation and subsequent H_2O_2 production are still a controversial issue. In *Drosophila*, Ca^{2+} appears to play an important role in the activation of Duox1 in vivo (7), although in vertebrates it was reported that Duox1 activation and Ca^{2+} waves are two independent early signals that appear after tissue injury (8). ATP and Ca^{2+} are two of the main molecules involved in early signaling after tissue damage (11–16). Therefore, both ATP and Ca^{2+} might be acting as upstream signals of Duox1. To study this, several pharmacological inhibitors were used to affect intracellular Ca^{2+} (TG and U73122) and purinergic signaling (PPADS,

suramin and apyrase) at different levels in wounded tail fins of zebrafish. The findings show that both extracellular ATP and intracellular Ca^{2+} have a key role in Duox1 activation/ H_2O_2 production in vivo and also in early neutrophil recruitment. PLC and purinergic receptor inhibition induced a significant decrease in both H_2O_2 release and neutrophil recruitment to the wound. Therefore, it is tempting to speculate that extracellular ATP acting through P2Y purinergic receptors might be triggering an increase in intracellular Ca^{2+} , which would then bind to the Duox1 EF hands, inducing its activation. This is consistent with previous in vitro studies that demonstrated that Duox1 activation depended on calcium binding to functional EF-hand motifs (49).

In zebrafish, by knocking-down Duox1 using a splice blocking MO, it was shown that Duox1 is the main NADPH oxidase responsible for the formation of the H_2O_2 tissue gradient in vivo (5, 6, 41, 48). The use of this splice blocking MO involves a very precise microinjection of the eggs, because excessively high MO doses generally compromise larvae viability, inducing several problems in larvae development that require coinjection of the p53-blocking MO to restore larvae viability (5). Moreover, in studies of signaling pathways the use of the p53 MO could interfere with the results, restricting the usefulness of this method to improve larvae viability. Because of these problems, it was felt necessary to develop another technique that would affect Duox1

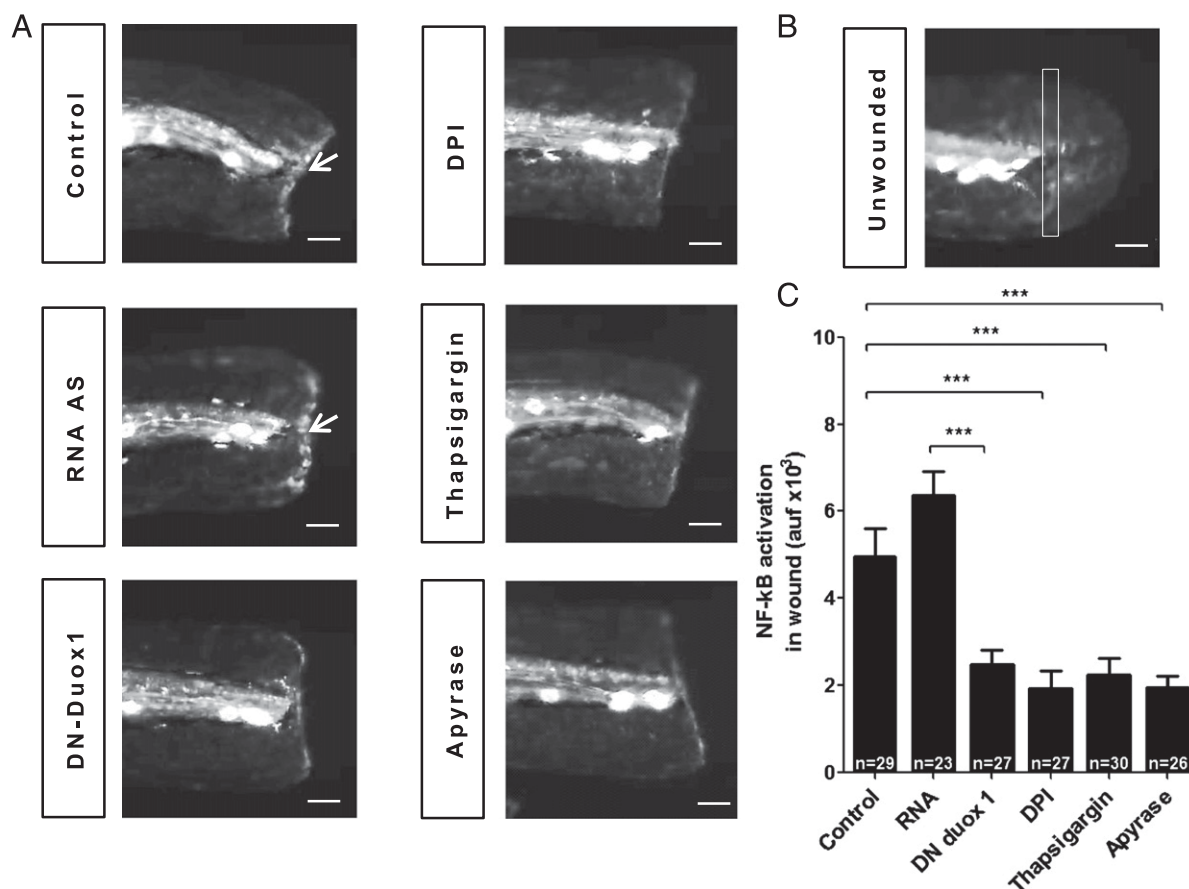


FIGURE 4. H_2O_2 activates the NF- κ B inflammatory signaling pathway. Zebrafish one-cell Tg(NF- κ B:eGFP) embryos were microinjected with antisense (RNA AS) or sense DN-Duox1 (DN-Duox1) mRNA. At 3 dpf, uninjected larvae were treated with TG or apyrase. Further tail fin larvae amputation was performed in all groups. (A) Representative maximum intensity projections of green channel of wounded larvae tail fins (arrows) at 30 minpw. (B) Representative maximum intensity projections of green channel of unwounded larvae tail fin. (C) Wound fluorescence intensity quantification of wounded larvae tail fins at 30 minpw. (Control: $n = 29$, RNA AS: $n = 23$, DN-Duox1: $n = 27$, DPI: $n = 27$, TG: $n = 30$, apyrase: $n = 26$). NF- κ B was activated in wounded larvae tail fin tissue. This signal suffers a significantly decrease in the presence of DN-Duox1, DPI, TG, or apyrase. All data are represented as means $\pm$ SEM. Scale bars, 100 μ m. The p values were calculated using one-way ANOVA and Bonferroni multiple comparison test (***) $p < 0.001$. auf, arbitrary unit of fluorescence.

activity to study the mechanisms involved in the generation of the tissue-scale H_2O_2 gradient. We hypothesized that a truncated form of Duox1 without the binding site for NADPH and FAD, the so-called flavin domain, would be able to compete with the native form of Duox1 and, therefore, act as a dominant negative form and decrease the H_2O_2 production, as has been found for NOX4 (42). Surprisingly, besides the negative aspects already reported for the use of Duox1 MOs (5, 6, 43), we found that they also induce an inflammatory process in 3 dpf larvae, which is responsible for the mobilization of neutrophils from the CHT to other tissues. We speculated that this phenotype could be due to the effect of this MO on larvae development and the induction of cell death. Importantly, the developed DN-Duox1 does not affect development or neutrophil distribution. Using tail fin amputation, we were able to demonstrate that the DN-Duox1 is functionally able to decrease in vivo H_2O_2 production and subsequent neutrophil recruitment. Importantly, this effect was specific, because neutrophil recruitment to LTB_4 was unaffected by DN-Duox1. Although further studies are required to understand the mechanism involved in the inhibition of endogenous Duox1 by the DN form lacking the flavin domain, it is tempting to speculate that the DN is competing with wild-type form for the Duox1 maturation factor, called Duox activator 1 (Duoxa1), which is important for Duox1 trafficking to the plasma membrane and its full function (50). The low expression of GFP-tagged DN-Duox1 in larvae is consistent with this hypothesis, because Duox1 would be expressed properly only in epithelial tissues. Whether or not this is the case, our results demonstrate that the DN-Duox1 is a useful tool for future studies in the field, because its use avoids the problems of the MOs that have been used until now (5, 6, 48).

Our epistasis study by ear injection of different stimuli combined with pharmacological inhibition of purinergic or Ca^{2+} signaling allowed us to establish whether these signals act in parallel or whether one is upstream of the other. We observed that all the stimuli, including H_2O_2 , ATP, ADP, and the calcium ionophore A23187, were able to induce neutrophil recruitment in control conditions. Furthermore, in the presence of the DN-Duox1, only the H_2O_2 was able to induce neutrophil recruitment to the ear cavity, demonstrating that both ATP and iCa^{2+} are acting as upstream signals of Duox1. We also found that ATP and calcium are not two independent signals and that ATP is actually an upstream signal of iCa^{2+} , because H_2O_2 and the Ca^{2+} ionophore were the only two stimuli able to significantly increase neutrophil recruitment in larvae pretreated with TG and the PLC inhibitor U73122. Taken together, these data strongly support the view that Duox1 activation involves metabotropic P2Y receptor signaling, rather than ionotropic P2X purinergic receptor activation, because P2X signaling is not triggered by ADP and does not involve PLC activation (51, 52).

NF- κB is a master regulator of inflammation and it controls the expression of several proinflammatory genes (47). Although there are still many inconsistencies concerning the influence of oxidative stress on NF- κB activity (53), several in vitro studies using H_2O_2 and cultured cells have shown that H_2O_2 can act as a regulator of I κB kinases (54–57). Our results demonstrate that the inhibition of Duox1-derived H_2O_2 production, by the direct genetic inhibition of Duox1 as a result of the overexpression of the DN-Duox1 or indirect pharmacological inhibition of upstream purinergic and Ca^{2+} signaling, decreased the NF- κB activation in wounded tissue. These data indicate, therefore, that NF- κB is

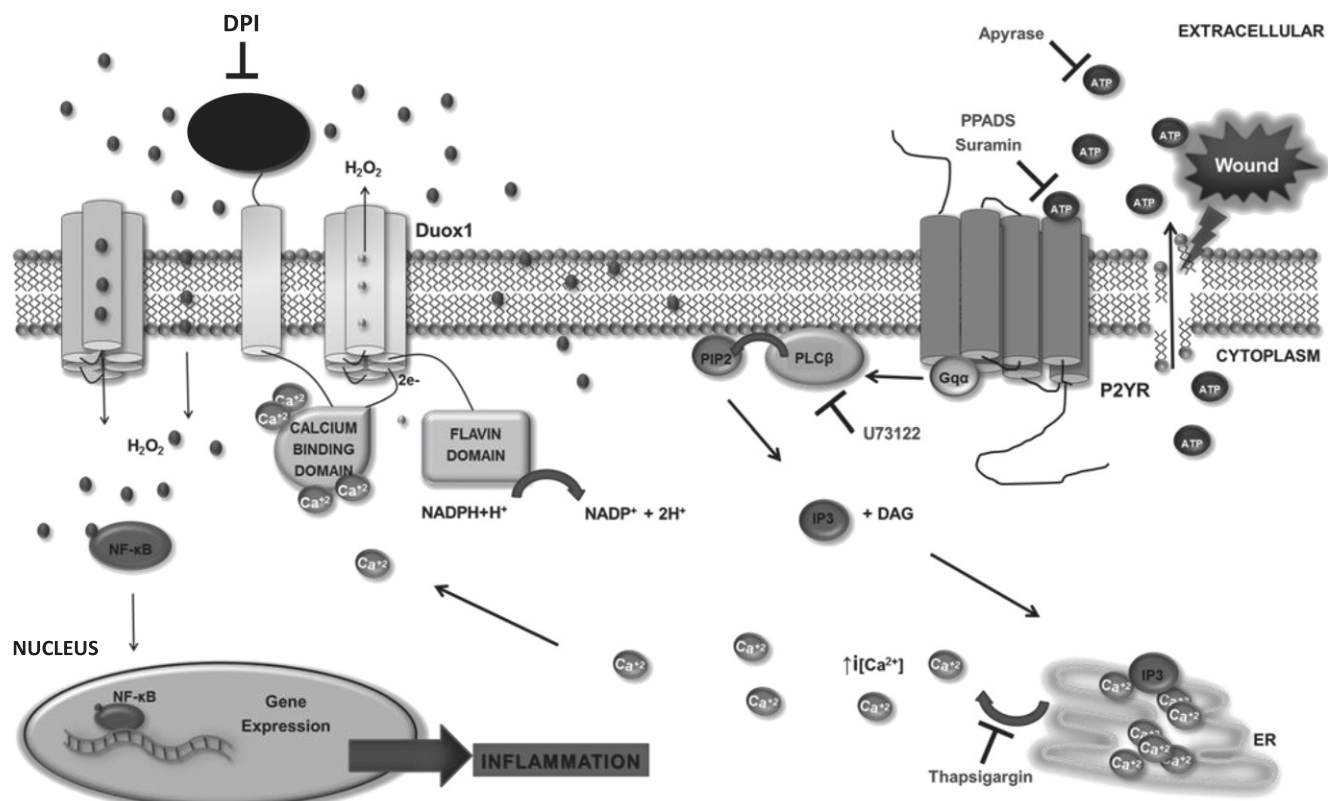


FIGURE 5. Proposed model illustrating how early signals (ATP, calcium and H_2O_2) modulate inflammation via NF- κB activation. ATP released from damaged cells activates purinergic P2YR, which promotes the activation of PLC, the generation of IP3 and the release of Ca^{2+} from the endoplasmic reticulum. Ca^{2+} is then able to bind the EF hands domain of Duox1, inducing its activation and the production of H_2O_2 . Finally, H_2O_2 is able to activate the master inflammation transcription factor NF- κB , which induces the expression of target proinflammatory genes. The sites of action of the different pharmacological drugs used in this study are also indicated.

a redox-sensitive transcription factor activated by H_2O_2 in vivo and indicate that the tissue-scale H_2O_2 gradient is a general player in the inflammatory process (Fig. 5).

In summary, we have developed a new and promising DN form of zebrafish Duox1 which is able to inhibit endogenous Duox1 activity, affecting tissue-scale H_2O_2 gradient formation and early neutrophil recruitment to wounds. Using this tool, we have demonstrated that ATP released at wound triggers Duox1 activation and local H_2O_2 release via a P2YR/PLC/ Ca^{2+} signaling pathway. Finally, we demonstrate for the first time that the early danger signals ATP, iCa^{2+} and H_2O_2 are necessary to activate in vivo the NF- κ B inflammatory signaling pathway, uncovering an unprecedented role for these early signals in the modulation of wound inflammatory responses. Our study identifies new potential therapeutic targets for inflammatory diseases and paves the way for future studies on the mechanisms orchestrating H_2O_2 oxidative regulation of complex biological processes, such as inflammation, regeneration, and cancer.

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Disclosures

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